

MARIAPFARR, Austria

WMO: 113480

Lat: 47.152N

Lon: 13.745E

Elev: 1153

StdP: 88.22

Time Zone: 1.00 (EUC)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-13.5	-18.0	0.9	-14.5	-16.0	1.1	-12.1	9.9	-2.7	8.7	-3.4	1.0	350	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	26.3	17.0	24.5	16.2	22.8	15.5	17.8	24.4	16.9	23.2	16.0	21.6	2.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	12.7	20.3	14.6	11.9	19.1	13.7	11.3	18.1	55.2	24.5	52.0	23.3	49.2	21.7	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.1	6.3	-18.8	28.9	3.3	1.5	-21.2	30.0	-23.1	30.9	-25.0	31.7	-27.4	32.8		
(5)			-19.0	19.3	3.3	1.2	-21.3	20.1	-23.2	20.8	-25.1	21.5	-27.4	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-4.0	-1.9	1.6	5.5	10.0	13.4	15.3	15.1	11.0	7.0	1.9	-2.9		
	DBStd	7.72	4.32	4.21	4.08	3.54	3.17	3.56	3.08	3.08	3.41	3.61	4.17	4.25		
	HDD10.0	2000	434	334	261	137	40	11	2	3	29	104	245	400		
	HDD18.3	4501	692	567	519	384	258	149	100	106	219	353	494	658		
	CDD10.0	556	0	0	0	4	41	114	167	161	59	10	1	0		
	CDD18.3	15	0	0	0	0	0	2	7	6	0	0	0	0		
	CDH23.3	288	0	0	0	0	7	62	113	102	3	0	0	0		
	CDH26.7	36	0	0	0	0	0	4	14	17	0	0	0	0		
(14) Wind	WSAvg	2.1	2.1	2.4	2.6	2.6	2.5	2.3	2.2	1.9	1.9	1.8	1.9	1.7		
(15) Precipitation	PrecAvg	1007	38	35	52	57	91	132	151	141	99	88	76	46		
	PrecMax	1190	76	120	113	94	131	213	264	223	161	214	177	145		
	PrecMin	828	7	5	11	21	46	72	26	85	56	16	0	7		
	PrecStd	112	23	27	25	18	23	35	56	34	27	42	43	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	10.5	15.6	19.1	24.3	27.2	28.5	28.7	23.7	19.5	14.2	9.7		
		MCWB	4.4	4.9	8.5	10.8	14.4	17.0	18.2	18.4	15.7	12.8	8.9	4.4		
	2%	DB	6.0	7.8	12.8	16.7	21.5	25.1	26.3	26.1	21.1	16.9	11.6	6.2		
		MCWB	2.9	3.5	6.5	9.7	13.4	16.0	17.3	17.2	14.3	11.1	7.8	3.3		
	5%	DB	4.1	5.8	10.6	14.8	19.4	23.2	24.3	24.0	19.2	14.8	9.7	4.3		
		MCWB	1.5	2.4	5.5	8.7	12.4	15.3	16.6	16.2	13.1	10.2	6.7	2.0		
	10%	DB	2.1	4.0	8.2	12.7	17.2	21.1	22.3	21.8	17.2	12.8	7.8	2.5		
		MCWB	0.4	1.3	4.4	7.4	11.2	14.4	15.7	15.3	12.4	9.4	5.8	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	5.4	8.7	11.5	15.4	18.1	19.2	19.2	16.5	13.5	9.7	5.6		
		MCDB	7.9	9.3	14.7	18.1	22.3	25.5	26.4	27.1	22.2	18.0	12.8	7.7		
	2%	WB	3.2	4.0	7.0	10.1	13.9	16.7	18.0	17.7	15.0	12.0	8.2	3.6		
		MCDB	5.4	7.1	11.8	15.7	20.5	23.4	24.2	24.5	19.9	15.5	10.7	5.7		
	5%	WB	1.7	2.6	5.7	9.0	12.8	15.8	17.1	16.7	13.9	10.9	7.2	2.3		
		MCDB	3.4	5.4	9.8	14.0	18.5	21.9	23.1	22.8	17.9	13.8	9.2	3.9		
	10%	WB	0.5	1.5	4.6	7.9	11.7	14.8	16.1	15.7	12.9	9.8	5.9	1.0		
		MCDB	2.1	3.7	7.9	11.9	16.4	20.1	21.4	20.9	16.3	12.3	7.8	2.3		
(35) Mean Daily Temperature Range	MDBR		7.8	8.7	9.5	10.0	10.9	10.7	10.9	11.0	10.1	10.2	7.2	7.1		
	5% DB	MCDBR	9.4	11.9	14.4	14.9	15.7	15.6	15.1	15.4	13.7	13.5	10.4	9.2		
		MCWBR	6.5	7.7	8.4	8.1	7.9	7.0	6.9	7.1	7.1	7.9	6.8	6.6		
	5% WB	MCDBR	8.4	10.7	13.3	13.7	14.7	14.1	14.0	14.4	12.0	11.9	8.9	8.4		
		MCWBR	6.3	7.3	7.9	7.6	7.6	6.6	6.7	6.8	6.5	7.2	6.1	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.297	0.343	0.388	0.399	0.405	0.405	0.397	0.363	0.335	0.302	0.269		
	taud		2.386	2.307	2.260	2.222	2.246	2.274	2.275	2.316	2.388	2.442	2.468	2.448		
	Ebn at Noon		833	869	874	864	865	860	854	846	845	818	781	790		
	Edh at Noon		69	93	114	131	133	130	128	118	100	81	65	58		
(44) All-Sky Solar Radiation	RadAvg		1.32	2.22	3.25	4.15	4.80	5.33	5.40	4.78	3.51	2.40	1.36	1.09		
	RadStd		0.16	0.29	0.39	0.51	0.56	0.48	0.47	0.54	0.45	0.26	0.23	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (96 neighbors)	+0.47	N/A	N/A	+0.76	+0.49	+0.38	N/A	-122	+81	+48

Nomenclature: See separate page